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Project Title: AI-Gamified Communication Channel for Enhancing Collaboration in Remote GitHub Teams

Domain of Interest: We focus on communication challenges in remote work settings, specifically within distributed software development teams using GitHub. This domain aligns with the AI's role in information technology organizations. It examines how AI can mitigate issues in asynchronous collaboration, such as fragmented discussion, reduced team engagement and motivation. These are common issues of remote team collaboration in global IT firms and open-source projects.

The problem we are trying to solve with this AI tool and why: From personal experience, GitHub projects are mostly text-based, asynchronous workflows, which leads to lurkers who might observe without contributing and lose follow-ups in threads. Many users have low motivation for tasks like code reviews. We are building this tool to introduce gamification and try to push routine interactions rewarding and fostering team cohesion in the GitHub community. This tool is proposed to make a practical AI solution that can improve remote team dynamics in IT organizations.

Target users: Developers and contributors in distributed GitHub teams, including lurker contributors who read issues and pull requests but seldom commit. Possible: a team lead who monitors project progress.

Project Plans: We plan to prototype a simple AI tool using low-code platforms. This tool is aimed to integrate GitHub APIs with a Slack or Discord channel to gamify interactions.

We plan to use low-code platforms like ChatGPT/Claude for actionable implementations (might) including elements:

The role of AI features include:	***Needed resources***:
(1) AI-generated personalized challenges based on user activity (e.g., "Review one PR to earn a "collaborator badge"). (2) Progress tracking with leaderboards, points, badges among all channel members. (3) (If applicable) AI-driven sentiment analysis for feedback on discussions in the channel.	(1) GitHub-Slack/Discord integration (2) Dashboard showing workflows and leaderboards (3) GitHub API for data pulls (issues/pull requests) (4) AI model API for AI features

The current timeline of this project:

- Week 5-6 (9/22-10/3): Setup and check all needed resources including API accessibility.
- Week 7 (10/6-10/10): Framework for UI design and implement personalization analysis. If time allows, do internal testing of basic functionality.
- Week 8 (see below goals): Internal testing and mid-semester check-in (on 10/16 dis sec2)
- Week 9-10 (10/20-10/31): Refine prototype.
- Week 11-14: Ethical auditing and prepare showcase (adjustable)
- Week 15: Final showcase

The final artifact will include the AI tool demo, an analysis of its organizational effects, and societal implications in real-world scenarios.

Challenges We May Encounter

- The first challenge will be technical: API limits, either GitHub or AI model, will slow real-time features.
- Potential biases in AI personalization might raise for some low-activity users.
- The ideal adoption testing is recruiting testers across time zones but it might be hard to call enough people in the short time.
- The balance between bias considerations and gamification impact

Goals for the Mid-Semester Check-In (Week 8)

- Demo with core features of the AI tool on a test repo integration.
- Evaluation of the internal testing.
- Showcase how the tool works (live or video record in advance).
- Future direction of iterations with the feedback.

Questions

- What demo formats are preferred for the showcase week?
- For the mid-semester demo, focus more on technical functionality, or on user experience and organizational impact?
- Would a small-scale user test, such as peer feedback or a simulated GitHub project, be sufficient, or is a more formal experiment expected?